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Amir Rafe, Ph.D.

Postdoctoral Researcher

Education

2022–2025 **Ph.D. in Civil & Environmental Engineering**, *Utah State University*, Transportation Engineering specialization, GPA: 4.0
Dissertation: EvacuAIDi: An AI-Driven, Causal-Informed Framework for Probabilistic and Disability-Inclusive Evacuation Guidance

2010–2013 **M.S. in Civil & Environmental Engineering**, *Azad University*, Transportation Engineering specialization, GPA: 4.0
Thesis: Pedestrian Dynamics Modeling Using the Social Force Model: A Heterogeneous Approach

2005–2009 **B.S. in Civil Engineering**, *Azad University*, GPA: 4.0

Research Interests

Traffic safety and vulnerable road users; causal and trustworthy AI for transportation systems; connected and automated infrastructure; commercial motor vehicle risk; pedestrian dynamics and evacuation modeling; quantum causal AI.

Academic and Professional Appointments

2025–Present **Postdoctoral Researcher**, *AI in Transportation (AIT) Lab, Texas State University, Ingram School of Engineering*

2022–2025 **Graduate Research Assistant**, *Singleton Transportation Lab, Utah State University, Department of Civil & Environmental Engineering*

2012–2022 **Algorithm Engineer**, *Ramona T&I and IUST Research Institutes*
Transportation modeling, simulation, software development, and applied traffic-safety studies.

Publications

Peer-Reviewed Journal Articles

2027 Gong, P., Lu, Y., Feng, Z., **Rafe, A.**, Shen, X., & Lovreglio, R. "Integrating Semantic and Operational Capabilities into IFC-Compliant BIM for Building Design: A Mixture-of-Experts Fine-Tuning Approach." *Automation in Construction*, 193, 107355. https://doi.org/10.1016/j.autcon.2026.107355.

2026 **Rafe, A.**, & Singleton, P. A. "Who Leads, Who Follows? A Review of Leadership and Guidance Dynamics in Evacuation." *Safety Science* (accepted).

Somvanshi, S., Islam, M. M., Mimi, M. S., Polock, S. B. B., Chhetri, G., Dutta, A., **Rafe, A.**, & Das, S. "Advancing Intelligent Sequence Modeling: Evolution, Trade-offs, and Applications of State-Space Architectures from S4 to Mamba—A Survey." *Neurocomputing* (accepted).

Rafe, A., & Das, S. "Do Aviation Hazard Rankings Overlook Conservation-Concern Birds in U.S. Wildlife Strikes?" *Transportation Research Part D: Transport and Environment*, 161, 105602. https://doi.org/10.1016/j.trd.2026.105602.

Rafe, A., Singleton, P. A., & Christensen, K. M. "Experimental Analysis and Disability-Aware Social Force Modeling of Heterogeneous Evacuation with Counterfactual Route Guidance." *Physica A: Statistical Mechanics and its Applications*, 131992. https://doi.org/10.1016/j.physa.2026.131992.

GBD 2023 Road Injuries Collaborators (**Rafe, A.**, named collaborator). "Global, regional, and national burden of road injuries 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023." *The Lancet Public Health*. https://doi.org/10.1016/S2468-2667(26)00141-6.

Barua, S., Starewich, M., Chowdhury, T. I., **Rafe, A.**, & Das, S. "Decoding Pedestrian Severity at Crosswalks using Hybrid Clustering and Random Parameter Models." *Scientific Reports*. https://doi.org/10.1038/s41598-026-56530-5.

2025 Somvanshi, S., Islam, M. M., **Rafe, A.**, Tusti, A. G., Chakraborty, A., Baitullah, A., Chowdhury, T. I., Alnawmasi, N., Dutta, A., & Das, S. "Bridging the Black Box: A Survey on Mechanistic Interpretability in AI." *ACM Computing Surveys*. https://doi.org/10.1145/3787104.

Rafe, A., Singleton, P. A., Boyer, S., & Mekker, M. "Pedestrian Crossing Behaviors at Signalized Intersections in Utah: Factors Affecting Spatial and Temporal Violations." *International Journal of Transportation Science and Technology*. https://doi.org/10.1016/j.ijtst.2024.12.005.

2024 **Rafe, A.**, Arman, M. A., & Singleton, P. A. "A Comparative Study Using Generalized Ordered Probit, Stacking Ensemble, and TabNet: Application to Determinants of Pedestrian Crash Severity." *Data Science for Transportation*, 6(2), 13. https://doi.org/10.1007/s42421-024-00098-x.

Rafe, A., & Singleton, P. A. "Imputing Time Series Pedestrian Volume Data With Consideration of Epidemiological-Environmental Variables." *Transportation Research Record*. https://doi.org/10.1177/03611981241240758.

Rafe, A., & Singleton, P. A. "Exploring the Complexity of Pedestrian Dynamics: Impact of Societal Behaviors and Personal Attributes in Urban Environments." *Transportation Research Record*. https://doi.org/10.1177/03611981241260707.

Books, Book Chapters, and Technical Reports

Das, S., & **Rafe, A.** *Agentic AI in Civil Engineering*. Elsevier (under contract).

2027 **Rafe, A.**, & Singleton, P. A. "Causal Inference for Active Transportation and Subjective Well-Being: Package Effects, Proxy-Augmented Learning, and Synthetic Validation." In M. Z. Naser (Ed.), *Causal Machine Learning in Civil and Environmental Engineering: Case Studies and Datasets* (Chapter 3). Elsevier. https://doi.org/10.1016/B978-0-443-15714-1.00004-7.

2026 Das, S., **Rafe, A.**, Mimi, M. S., Barua, S., & Islam, M. M. "Traffic Analysis Practices for Nonmotorized Modes." *NCHRP Synthesis 671*, Transportation Research Board. https://doi.org/10.17226/29551.

Conference Proceedings and Presentations (selected)

2027 **Rafe, A.**, & Das, S. "Human Versus Machine Braking Decisions in Automation-Engaged Rear-End Crashes Mined from Statewide Police Reports." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-02821).

Rafe, A., & Das, S. "Auditable Causal Mental Models from Crash Narratives for Unprotected Road Users." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-04913).

Rafe, A., & Das, S. "Auditable Coverage Guarantees for Deployed Crash Injury Severity Models." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-03247).

Rafe, A., & Das, S. "Do Language Models Invent the Safety Data They Cannot Find? A Multi-Model Benchmark of Large Language Model Abstention and Fabrication in Injury-Narrative Extraction." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-05035).

Rafe, A., Ghazaryan, A., Amaro, E., & Ronchi, E. "Can a Vision-Language Model Time Evacuation? Validating Automated Video Annotation of Response Behavior Against Game-Engine Ground Truth." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-06259).

Rafe, A., Yadav, R. K., & Das, S. "How Hard Does the World Cup Hit the Road Network? A Cross-City Resilience Analysis of FIFA 2026 Knockout Matches Using High-Resolution Probe Data." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-01996).

Rafe, A., & Das, S. "How Many Vehicle Defects Have Already Occurred but Not Yet Been Reported? Nowcasting the NHTSA Consumer Complaint Stream with Reporting-Delay Correction." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-03187).

Shuvo, S. A., Barua, S., Chowdhury, S. S. U., **Rafe, A.**, & Das, S. "Presumed Intended and Unintended Passing-Related Crashes at No-Passing Zones in Texas: A Narrative-Informed Association Rule Mining Analysis." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-02971).

Yadav, R. K., **Rafe, A.**, Chhetri, G., & Das, S. "Do Not Prune the Crash: Criticality-Aware Compute Allocation with Conformal Rare-Event Recall Control for Streaming Transportation Video-VLMs." *106th TRB Annual Meeting*, Washington, D.C. (TRBAM-27-06122).

2026 Das, S., & **Rafe, A.** "Causal-Safe: A Multimodal Neuro-Symbolic Agent for Counterfactual Traffic Risk Assessment and Intervention." Poster, *TechConnect World Innovation Conference & Expo*, Raleigh, NC.

Das, S., & **Rafe, A.** "A Privacy-Preserving Cyberinfrastructure for Synthetic Transportation Video and Safety Analytics." Poster, *TechConnect World Innovation Conference & Expo*, Raleigh, NC.

Das, S., & **Rafe, A.** "A Sheaf-Theoretic Framework for Trustworthy Multi-Sensor Fusion in Roadway Safety." Poster, *TechConnect World Innovation Conference & Expo*, Raleigh, NC.

2025 **Rafe, A.**, Lawrence, P. J., Lovreglio, R., Spearpoint, M., & Singleton, P. A. "Enhancing Occupant Evacuation Simulation Using LLMs and Retrieval-Augmented Generation." *ASCE ICTD 2025*, Glendale, AZ. https://doi.org/10.1061/9780784486191.034.

Afand, K., **Rafe, A.**, Khademi, N., Mazloum, S., Ahmadi, A., Zabihpour, A., & Singleton, P. A. "Virtual Reality Analysis of Leadership Dynamics and Pedestrian Route Choice During Mass Gathering Evacuations." *ASCE ICTD 2025*, Glendale, AZ. https://doi.org/10.1061/9780784486207.004.

Mazloum, S., Khademi, N., Ahmadi, A., Afand, K., Zabihpour, A., **Rafe, A.**, & Singleton, P. A. "Exploring Pedestrian Evacuation Behavior Using a Virtual Reality Environment: An Eye-Tracking Study." *ASCE ICTD 2025*, Glendale, AZ. https://doi.org/10.1061/9780784486207.035.

2024 **Rafe, A.**, & Singleton, P. A. "Exploring the Determinants of Pedestrian Crash Severity Using an AutoML Approach." *ASCE ICTD 2024*, Atlanta, GA, 442–455. https://doi.org/10.1061/9780784485514.039.

2023 **Rafe, A.**, Kretz, T., & Singleton, P. A. "Importance of Social Behaviors on Pedestrian Dynamics: A Case Study of Islamic Clothing in Iran." *102nd TRB Annual Meeting*, Washington, D.C.

2019 Arman, M. A., **Rafe, A.**, & Kretz, T. "Applied Hybrid Binary Mixed Logit to Investigate Pedestrian Crossing Safety at Midblock and Unsignalized Intersection." arXiv. https://doi.org/10.48550/arXiv.1905.09403.

2015 Arman, M. A., **Rafe, A.**, & Kretz, T. "Pedestrian Gap Acceptance Behavior, A Case Study: Tehran." *94th TRB Annual Meeting*, Washington, D.C.

2014 **Rafe, A.**, & Karimi, M. "Calibrating Social Force Model Based on Design Experiments Method." *13th International Conference on Traffic and Transportation Engineering.*

2014 **Rafe, A.**, & Khavarzade, R. "Investigation of Pedestrian Gap Acceptance Behavior at Crosswalk." *13th International Conference on Traffic and Transportation Engineering.*

2014 **Rafe, A.**, Shariat, A., Kalantari, N., Arman, M. A., & Yazdanpanah. "Pedestrian Simulation with Viswalk at Various Walking Facilities." *13th International Conference on Traffic and Transportation Engineering.*

Manuscripts Under Review and Preprints (selected)

2026 **Rafe, A.**, & Das, S. "Estimating Uncoded Crash Factors with Tabular Foundation and System One Models: Kumo Tabular and Jev." Submitted to *ACM Transactions on Intelligent Systems and Technology.* https://doi.org/10.48550/arXiv.2610.10321.

Rafe, A., & Das, S. "Benchmarking System One Decision Models Against Trained Classifiers and Language Models for Automated Decision Gates." arXiv preprint. https://doi.org/10.48550/arXiv.2610.00346.

Rafe, A., & Das, S. "Counting the Uncounted: Population-Level Surveillance of Documented Pregnancy and Fetal Harm in Police Crash Narratives with a System One Model (Jev)." Submitted to *Accident Analysis & Prevention.* https://doi.org/10.48550/arXiv.2610.00213.

Rafe, A., & Das, S. "Calibrated Decisions at Scale: Converting Police Crash Narratives into Probabilistic Crash Variables with a System One Model (Jev)." Submitted to *Accident Analysis & Prevention.* https://doi.org/10.48550/arXiv.2609.24052.

Rafe, A., & Das, S. "Narrative-Validated Roles of Unprotected Road Users in a Decade of California Automated Vehicle Collisions." Submitted to *Accident Analysis & Prevention.*

Rafe, A., & Das, S. "Differential Reporting and Recovered Behavioural Mechanisms in Electric Scooter Injury Surveillance Under a Safe Systems Lens." Submitted to *Safety Science.*

Rafe, A., Baitullah, A., & Das, S. "Latent Profiles of AI Risk Perception and Their Differential Association with Community Driving Safety Concerns: A Person-Centered Analysis." Submitted to *Transportation Research Part F: Traffic Psychology and Behaviour.* https://doi.org/10.48550/arXiv.2604.04849.

Rafe, A., & Das, S. "Heterogeneous Ordinal Structure Learning with Bayesian Nonparametric Complexity Discovery." arXiv preprint. https://doi.org/10.48550/arXiv.2605.04191.

Rafe, A., & Das, S. "Socio-Conformal Calibration in Complex Survey Data: Marginal Validity Is Not Enough for Subgroup Reliability." arXiv preprint. https://doi.org/10.48550/arXiv.2605.05562.

2025 **Rafe, A.**, & Das, S. "Causal AI and Computational Epistemology: A Survey." Submitted to *ACM Computing Surveys*. https://dx.doi.org/10.2139/ssrn.6045495.

Rafe, A., & Singleton, P. A. "Causal AI for Evacuation Modeling: A Modular Framework for Standards-Compliant Simulation Input Generation." Submitted to *Transportation Research Part D: Transport and Environment*. Included in doctoral thesis.

Rafe, A., & Singleton, P. A. "Evaluating AI-Driven Evacuation Guidance: A Causal-Intervention and Counterfactual Simulation Framework." Submitted to *Transportation Research Part C: Emerging Technologies*. Included in doctoral thesis.

Rafe, A., & Singleton, P. A. "A Hierarchical Bayesian Framework for AI-Guided Evacuations under Human Compliance Uncertainty." Submitted to *Transportation Research Part C: Emerging Technologies*. Included in doctoral thesis.

Research Projects and Funding Experience (selected)

2026–Present FMCSA: *Telematics-Informed Crash Risk Modeling to Support Targeted Traffic Enforcement for CMV Safety*. Sponsor: Federal Motor Carrier Safety Administration. Role: Key Researcher.

CREATE University Transportation Center: *Role of Emerging Transportation Technologies and Safety Initiatives in Mitigating Crashes in Coastal Communities*. Role: Senior Personnel.

2025–Present NCHRP 17-113: *Incorporating Safe System Approach into the NCHRP Report 500 Series*. Sponsor: National Cooperative Highway Research Program. Role: Key Researcher.

NCHRP Synthesis 671 (Topic 56-05): *Traffic Analysis Practices for Nonmotorized Modes*. Sponsor: National Cooperative Highway Research Program. Role: Key Researcher.

NCHRP 17-134: *Center Line Buffer Areas for Safety: Implementation Guidelines and Tool*. Sponsor: National Cooperative Highway Research Program. Role: Key Researcher.

BTSCRP BTS-39: *Toolkit for Reducing Substance-Impaired Driving for the Last Leg of the Journey*. Sponsor: Behavioral Traffic Safety Cooperative Research Program. Role: Key Researcher.

BTSCRP BTS-42: *Guidelines for Authorizing, Implementing, and Operating Automated Traffic Enforcement Programs*. Sponsor: Behavioral Traffic Safety Cooperative Research Program. Role: Key Researcher.

Examining the Pre-Crash Circumstances Leading to Pedestrian Fatalities. Sponsor: AAA Foundation for Traffic Safety. Role: Key Researcher.

2024 *AI-Powered Tools for Safe Evacuation of Individuals During Emergencies*. Sponsor: Center for Transformative Infrastructure Preservation and Sustainability (CTIPS). Role: Key Researcher.

2022–2024 *The UDOT Pedestrian Data Platform*. Sponsor: Utah Department of Transportation. Role: Key Researcher.

2022 *Prediction of Traffic Congestion Patterns Using Deep Learning*, in cooperation with KU Leuven. Role: Project Manager.

2021 *Fire Simulation and Crowd Emergency Evacuation Studies of Two Regional Airports*. Role: Project Manager.

2019–2021 *Urban Freight Master Plans* for two cities and an *Urban Transportation Master Plan* for a third. Sponsor: City Municipalities. Role: Project Manager.

2017–2020 *Traffic Safety and Impact Studies* for four cities. Sponsor: City Municipalities. Role: Project Manager.

2013–2016 Tunnel evacuation, pedestrian simulation, and guide-sign software. Role: Project Manager.

Proposals Written and Submitted (selected)

2026 NSF: *AIMing: Neuro-Symbolic Verification for the Autonomous Energy-Mobility Nexus*, Artificial Intelligence, Formal Methods, and Mathematical Reasoning (AIMing).

Amazon Research Awards: *RoadSafe Agents: Governed Multi-Agent AI for Proactive Road Safety Scoring.*

Amazon Research Awards: *SafetyLens and SafeLM: Risk-Calibrated Multimodal Foundation Models for Auditable Traffic Crash Understanding.*

2025 NSF: *Sheaf-Theoretic Identifiability for Multi-Sensor Safety Systems*, Mathematical Foundations of Artificial Intelligence (MFAI).

NSF: *PIVOT: Privacy-Preserving Infrastructure for Video Synthesis and Analysis in Open Transportation Safety*, Cyberinfrastructure for Sustained Scientific Innovation (CSSI).

ARPA-I: *EvacuVerse: A National Causal Digital Twin for Disability-Inclusive Evacuation and Multi-Modal Re-Routing.*

Patent Applications and Invention Disclosures

2026 **Rafe, A.**, & Das, S. "CausalSafe: System and Method for Multimodal Neuro-Symbolic Causal Inference, Counterfactual Traffic Risk Assessment, and Proactive Safety Intervention Using Structural Causal Models and Grounded Large Language Models." U.S. Provisional Patent Application No. 64/072,601, filed July 10, 2026. Filed through Texas State University.

Rafe, A., & Das, S. "PIVOT: System and Method for Privacy-Preserving Synthesis, Co-Reported Privacy-Utility Certification, and Federated Sharing of Transportation Video, Trajectories, and Safety Analytics." U.S. Provisional Patent Application No. 64/072,290, filed May 22, 2026. Filed through Texas State University.

Rafe, A., & Das, S. "Sheaf-Theoretic System and Method for Trustworthy Multi-Sensor Fusion in Roadway Safety Applications." U.S. Provisional Patent Application No. 64/072,334, filed May 22, 2026. Filed through Texas State University.

EvacuAIDi: Accessibility-aware AI framework for probabilistic emergency evacuation guidance. *Innovation disclosure submitted*, Utah State University.

Teaching Experience

2026 Guest Lecture: "System 1 Models in Safety Research: What Jev and Laya Can and Cannot Do," *AI in Civil Engineering* (CE 7393), Guest Speaker Series, Texas State University.

2025–2026 Guest Lecturer: *Traffic Engineering*; *AI in Civil Engineering* ("Agentic AI in Civil Engineering"); *Roadway Infrastructure Safety*, Texas State University.

2022–2024 Guest Lecturer: *Transportation Planning*; *Transportation Data and Safety Analysis*, Utah State University.

2023 Curriculum Developer, *Empowering Teaching Excellence (ETE-10)*: Prepared asynchronous modules for undergraduate and graduate students in statistics, data science, and applied AI.

Mentoring

Texas State University ***Ph.D.***: Sawgat Ahmed Shuvo. ***M.S.***: Arka Chakraborty; Anika Baitullah; Sazzad Bin Bashar Polock; Tausif Islam Chowdhury. ***B.S.***: Biplov Pandey; Irfan Sarwar Pranjol; Bidisha Shrestha. ***B.S. Computer Science***: Gaurab Chhetri; Radha Yadav.

Lund University ***M.S. co-supervisor***: Yousef Zoghi.

Academic and Professional Service

Conference Review TRB Annual Meeting, 2022–Present; NeurIPS, 2026; IEEE Intelligent Transportation Systems Conference (ITSC), 2024.

Journal Review Safety Science; Transportation Research Parts A, D, and F; Accident Analysis & Prevention; European Transport Research Review; International Journal of Transportation Science and Technology; Journal of Transport & Health; Journal of Transport Geography; Transportation Research Record; Transportation Research Interdisciplinary Perspectives.

Awards and Certifications

Award The BobCatalyst Innovation Accelerator Program (BIAP) Award (Texas State University); Graduate Enhancement Scholarship (Utah State University).

Software and Computing

Computing: Python; R; SQL (PostgreSQL, Neo4j, DuckDB, BigQuery); PyTorch; TensorFlow; scikit-learn; Hugging Face; LangChain; PyMC; DoWhy; Apache Spark; Docker; GCP; AWS.

Transportation: PTV Vissim/Viswalk; Synchro; SUMO; MATSim; QGIS; ArcGIS; Pathfinder; MassMotion; Fire Dynamics Simulator (FDS); AutoCAD Civil 3D.

Memberships and Languages

Memberships: American Society of Civil Engineers (ASCE); Institute of Transportation Engineers (ITE); Global Burden of Disease Collaborator Network (IHME).

Languages: English, fluent; Persian, fluent; Spanish, moderate.

References

- **Dr. Subasish Das**, Associate Professor, Ingram School of Engineering, Texas State University. Email: subasish@txstate.edu; Phone: +1-512-245-1826.
- **Dr. Patrick Singleton**, Associate Professor, Civil and Environmental Engineering, Utah State University. Email: patrick.singleton@usu.edu; Phone: +1-435-797-7109.
- **Dr. Ruggiero Lovreglio**, Professor in Digital Built Environment, School of Built Environment, Massey University. Email: R.Lovreglio@massey.ac.nz; Phone: +64-9-212-7114.